Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/25/2005
PI Name:	Barmatz, Martin Ph.D.		
Project Title:	Microgravity Test of Universality and Scaling Predictions Near the Liquid-Gas Critical Point of 3He		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rudnick, Joseph (University of California) Hahn, Inseob (Jet Propulsion Laboratory) Ulf, Israelsson (Jet Propulsion Laboratory) Zappoli, Bernard (Toulouse Space Centre)		
Grant/Contract No.:	None		
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Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	Scaling theories describing the behavior of thermophysical properties near critical points were developed using the homogeneity postulate. These models led to the definition of universality classes where critical points of the same class are predicted to have the same critical exponents. Efforts to test these models near a liquid-gas critical point in ground-based laboratories are limited due to the gravity-induced vertical density gradient associated with the divergence of the isothermal compressibility. This density gradient becomes appreciable as the critical point is approached, leading to a significant smearing of the transition. Calculations have shown that in a microgravity environment accurate, gravity free, heat capacity and isothermal susceptibility measurements could be obtained two orders of magnitude in reduced temperature closer to the critical point. Ground-based high-precision measurements performed in the gravity-affected as well as the gravity-free regions can also be used to test recent predictions of crossover equation-of-state models. High-resolution sensors are required for the temperature, pressure, and density measurements. The objectives of this task are to test the scaling equation-of-state predictions at the liquid-gas critical point of 3He. The task objectives will include: 1) Measurements of the heat capacity at constant volume along the critical isochore above and below the transition, and measurements of the isothermal susceptibility along the critical isochore above the transition and along the coexistence curve below the transition to determine the critical amplitudes and exponents associated with these diverging thermodynamic quantities. 2) Measurements of the density dependence of the pressure along the critical isotherm to determine the asymptotic critical amplitude and exponent for this diverging quantity. 3) Measurements along the coexistence curve to determine the asymptotic critical amplitudes and exponents associated with the isothermal susceptibility and the shape of the coexistence curve. 4) A rigorous test of equation-of-state predictions both in the asymptotic and crossover regions. The ability to perform these simultaneous measurements should permit a more accurate determination of the asymptotic critical exponents and amplitudes and test asymptotic equation-of-state predictions. Furthermore, knowledge of the asymptotic amplitudes (and exponents) will permit more stringent tests of crossover theories that apply farther away from the critical point. Critical phenomena exist not only in liquid-gas systems, but also in many other real-world systems such as magnetic systems, binary alloys, superconductors and superfluids. In fact the same scaling theory, being tested by MISTE, could apply to many seemingly different physical systems. Thus a high-precision examination of the scaling theory from MISTE constitutes a fundamental test of our basic understanding of material behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The unambiguous validation (or modification) of asymptotic and crossover equations-of-state models that apply to all simple fluids will provide a major advancement in the field of condensed matter physics.
Task Progress:	This last year has seen the publication of several articles of experimental and theoretical studies related to the MISTE experiment. A theoretical analysis of 3He critical point measurements titled “Comparison of Theoretical Models of Crossover Behavior near the 3He Liquid-Vapor Critical Point” was published in Physical Review E. The three advanced crossover models tested where the minimal subtraction renormalization model (MSR) (Dohm and co-workers), massive renormalization model (Bagnuls and Bervillier), and the crossover parametric model (CPM) (Anisimov and co-workers). The first two theoretical models applied renormalization group (RG) techniques to go beyond the asymptotic regime into the crossover regime. The third model, CPM, utilized an empirical crossover function that matched the known results from the RG theory in the asymptotic regime and from the mean field theory far away from the critical point. All three theoretical models provide the parametric expressions to calculate the isothermal susceptibility, specific heat, and coexistence curve. All the ratios of the leading critical amplitudes and the correction amplitudes were defined in these models. The 3He experimental data of isothermal susceptibility, specific heat, and coexistence curve were fit to all three theoretical models. The agreement between these theories and experimental measurements is good. The advantage of the CPM model over the first two is that it is a complete equation of state while the first two are only valid along the paths of the critical isochore and the coexistence curve. This is the first time that the CPM model has been rigorously tested against all three experimental properties simultaneously. We have also applied the CPM equation of state to calculate various experimentally measured quantities in the critical region close to the transition where the gravity-induced density stratification is significant. We have also re-analyzed the high precision microgravity heat capacity measurements in SF6 using the minimal subtraction renormalization (MSR) model. This paper titled “Reanalysis of Microgravity Heat Capacity Measurements Near the SF6 Liquid-Gas Critical Point,” was published in the Int. J. of Thermophys. This re-analysis revealed an unexpectedly large asymptotic region that appears not to be consistent with renormalization predictions. Because of several journal editorial errors, an Erratum has been submitted for publication in the Int. J. of Thermophys. A major issue regarding the proposed design of the MISTE flight cell was an uncertainty in the expected thermal time constant as the critical point was approached. A theoretical analysis of the 2-D piston effect has now given sufficiently accurate predictions of the relaxation time. This work described in a manuscript titled “Temperature and Density Relaxation Close to the Liquid-Gas Critical Point: an Analytical Solution for Cylindrical Cells” was published in Physical Review E. A recent analysis of earlier thermodynamic measurements near the 3He liquid-gas critical (Anisimov, M.A., F. Zhong, and M. Barmatz, “Resolving the Yang-Yang Dilemma in 3He Near the Critical Point,” J. Low Temp. Phys. 137, 69 (2004)) revealed subtle gravity effects (previously ignored) that require the careful placement of the pressure sensor within the experimental cell. The existing pressure sensor has been repositioned within the experimental cell to minimize gravity effects. We have redesigned and fabricated a new type of miniaturized high-resolution thermometer, using the metalized paramagnetic compound of PdMn. This new type of HRT has a fast equilibration and less susceptible to moisture contamination than an original GdCl3 salt type. We have also redesigned and fabricated a new capacitive density sensor using copper coated sapphire disks. The new density sensor has much improved parallelism between two electrodes. It also conducts away any electrical field induced heat within the capacitor gap much faster than previous designs. The updated MISTE low temperature probe is now going through calibration tests in preparation for thermodynamic measurements throughout the critical region. A new probe is also being fabricated to provide better thermal isolation in future planned experiments. An advanced data analysis approach that includes the calculation of confidence contours has also been developed to compare our thermodynamic measurements to theory. That approach was recently used in the reanalysis of SF6 microgravity specific heat measurements.

Bibliography Type:	Description: (Last Updated: 08/04/2008)
Articles in Peer-reviewed Journals	Barmatz M, Zhong F, Shih A. "Reanalysis of microgravity heat capacity measurements near the SF6 liquid-gas critical point." International Journal. of Thermophysics. 2004 Nov;25(6):1667-75. , Nov-2004
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Articles in Peer-reviewed Journals	Carles P, Zhong F, Weilert M, Barmatz M. " Temperature and density relaxation close to the liquid-gas critical point: an analytical solution for cylindrical cells." Phys Rev E Stat Nonlin Soft Matter Phys. 2005 Apr;71(4 Pt 1):041201. Epub 2005 Apr 4. PMID: 15903658 , Apr-2005
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